

Qwen3-8B SmartShop Fine-Tuning

Part 2: Instruction Tuning for Planner Accuracy

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Why Fine-Tune?

- General-purpose LLMs struggle to consistently output structured **JSON plans** matching exact API schemas.
- Hallucinations in tool names or missing dependency links break execution.
- Fine-tuning adapts a base model (**Qwen3-8B**) to:
 - Generate valid JSON plans under schema constraints.
 - Correctly invoke domain-specific e-commerce capabilities.
 - Parse subjective descriptions (e.g. colors, synonyms) into structured SQL lookups.

The Dataset Format

- Main training file: data/finetune_dataset.jsonl (1167 augmented instances).
- Structure: Instruction-Response JSON pairs.

Sample Dataset Item:

```
{
  "instruction": "I want to buy a blue Kurti readymade suit",
  "response": {
    "capabilities": ["search_products"],
    "parameters": {
      "category": "Suit",
      "color": "blue"
    }
  }
}
```

Training Script & Execution

- Script: `scripts/finetune_qwen3_8b_a100.py`
- Utilizes **Unsloth** and **QLoRA** on a Lightning AI A100 GPU for fast training.

Execution Command:

```
python scripts/finetune_qwen3_8b_a100.py \  
  --model-id unsloth/Qwen3-8B \  
  --dataset data/finetune_dataset.jsonl \  
  --output-dir models/qwen3-8b-smartshop-lora \  
  --max-seq-length 2048 \  
  --epochs 5 --batch-size 4 --grad-accum 2 \  
  --learning-rate 2e-4 --lora-r 16 --lora-alpha 32
```

Serving Architecture

- The trained adapter is packaged as `models/qwen3-8b-smartshop-lora.zip`.
- Served on a GPU server using **vLLM** with active LoRA support:

```
vllm serve Qwen/Qwen3-8B-AWQ \  
  --quantization awq --enable-lora \  
  --max-lora-rank 16 \  
  --lora-modules smartshop=/path/to/lora_dir \  
  --port 8000
```

- An **ngrok** HTTPS tunnel bridges the local GPU endpoint to the public web API.